

- Focusing on **Neati Krousa (Family & Community Circles)**
- Version 1 of NakPom MVP (Minimum Viable Product - Basic App - Version ដំបូងនៃapp)
- Focusing on one solid pillar ensures that we build:
 - A highly polished
 - Bug-free
 - Truly secure core
- The revised, hyper-focused **NakPom (អ្នកភូមិ)** project plan.
- All other pillars are moved to the Future Roadmap.
- The 4-week sprint has been rewritten to focus entirely on building the family network and community board.

NakPom (អ្នកភូមិ) V1 MVP Project Plan

1. Executive Summary & Vision

NakPom is a hyper-local social ecosystem designed to strengthen family and community bonds in Cambodia. Version 1 (V1 MVP) focuses exclusively on providing zero-friction, highly secure private digital spaces for "**Krousa**" (**family**) while seamlessly keeping citizens informed of local neighborhood updates. By delivering a Khmer-first, accessible interface with robust backend data isolation, NakPom ensures generational connectivity at absolutely zero infrastructure cost to the development team.

2. Core Feature Scope (V1 MVP vs. Future Roadmap)

I. The V1 MVP Focus: Neati Krousa (Family & Community Circles)

This module forms the absolute heart of the application, establishing a safe digital environment across generations.

- **"Krousa Me" (អ្នកផ្ទះខ្ញុំ) Default Sandbox:** The instant a user registers, the system automatically spins up a default private family group named "Krousa Me." The user can immediately test posting photos and text updates in this sandbox without being forced through a tedious onboarding setup.
- **Dynamic Invite Code System:** To expand the family space, a user can generate a short, alphanumeric code (e.g., NP-7892) with one tap and share it via Telegram or Messenger. Relatives paste this code on their devices to instantly join the private family circle.
- **The Private Family Feed:** A clean, secure timeline where authorized family members can share daily life updates, photos, and drop quick text messages.
- **Interactive Reactions:** High-contrast, easy-to-tap visual reaction buttons (like hearts or thumbs-up) so that less tech-literate family members can interact without heavy typing.
- **The Local Community Notice Board:** A secondary public-facing timeline within the app that displays localized, critical announcements or safety updates filtered by the user's district or commune level.

Future Roadmap (Subsequent Versions)

All advanced modules and secondary pillars are deferred to future implementation phases once the core family ecosystem is stable:

- **Neati Krousa Expansions:**
 - *Elderly Mode:* Voice-to-text Khmer input and highly simplified gesture navigation.
 - *AI Tools:* Built-in smart grammar correction and text autocomplete for Khmer script.
 - *Family Utilities:* Shared health histories, family calendar event management, and joint expense tracking.
- **Neati Sarpheap (Anonymous Voice & Community Reporting):** Map-based infrastructure incident reporting (Smart Sangkat), anonymous educational feedback walls (EduLink), and backend priority algorithms to rank community issues.
- **Neati Chamness (Knowledge & Skill Hub):** PDF scanning with AI summaries, video hosting pipelines for teachers, and a peer-to-peer matching API for local skill-swapping.
- **Neati Chumroeurn (Citizen Engagement & Surveys):** Micro-polling systems and analytical data-aggregation dashboards for local government bodies and NGOs.

3. Technical Stack & Data Architecture

Core Technologies

- **Frontend:** Jetpack Compose (Native Android) – organized for a modular, clean layout emphasizing a Khmer-first user experience.
- **Backend:** Kotlin (using Ktor or Spring Boot) – lightweight, asynchronous, and deployed on Linux environments for maximum cost-efficiency.
- **Authentication Flow:** Secure Email + Password registration. Passwords are salted and cryptographically hashed (using BCrypt). Password resets are handled securely and completely for free using transactional SMTP services (such as *Resend* or *Brevo*) to dispatch time-sensitive reset tokens directly to users' inboxes.

V1 Normal DB Schema Layout (MySQL)

The relational database structure is designed to enforce strict privacy walls so that family data remains entirely confidential:

```
SQL
-- 1. Core Users Table
CREATE TABLE users (
  user_id INT AUTO INCREMENT PRIMARY KEY
  email VARCHAR 100 NOT NULL UNIQUE
```

```

password_hash VARCHAR 255 NOT NULL
full_name VARCHAR 100 NOT NULL
created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

```

-- 2. Family Circles (Handles both "Krousa Me" and Joined Extended Spaces)

```

CREATE TABLE families
  family_id INT AUTO INCREMENT PRIMARY KEY
  family_name VARCHAR 100 NOT NULL
  invite_code VARCHAR 10 NOT NULL UNIQUE
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP

```

-- 3. Many-to-Many Membership Link (The Security Wall)

```

CREATE TABLE family_memberships
  membership_id INT AUTO INCREMENT PRIMARY KEY
  user_id INT
  family_id INT
  role VARCHAR 20 DEFAULT 'member' -- 'owner' / 'member'
  FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE
  FOREIGN KEY (family_id) REFERENCES families(family_id) ON DELETE CASCADE

```

-- 4. Secure Password Resets (Free SMTP Token Infrastructure)

```

CREATE TABLE password_resets
  id INT AUTO INCREMENT PRIMARY KEY
  email VARCHAR 100 NOT NULL
  token VARCHAR 255 NOT NULL UNIQUE
  expires_at TIMESTAMP NOT NULL
  INDEX(token) -- Optimized for rapid token lookup performance

```

-- 5. Private Posts Table

```

CREATE TABLE family_posts
  post_id INT AUTO INCREMENT PRIMARY KEY
  family_id INT
  user_id INT
  content_text TEXT
  image_url VARCHAR 255
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
  FOREIGN KEY (family_id) REFERENCES families(family_id) ON DELETE CASCADE
  FOREIGN KEY (user_id) REFERENCES users(user_id) ON DELETE CASCADE

```

4. Hyper-Focused 4-Week Agile Roadmap

Sprint 1: Identity & The "Krousa Me" Automator (Week 1)

- **Backend:** Set up the Kotlin development environment. Write data schemas and validation logic for User Registration and Login. Build the registration hook that automatically inserts a new family record named "Krousa Me" and links the user to it via family_memberships immediately upon signup.
- **Database:** Initialize the MySQL instance with indexes optimized for lookup tokens.
- **UML Deliverables:** Complete the standard Use Case and Class diagrams, mapping out these exact baseline entities.

Sprint 2: The Invite Engine & UI Scaffold (Week 2)

- **Backend:** Develop the invite logic endpoints (GET /api/v1/family/join?code=...). Build security filters that intercept feed requests to verify that the requesting user_id holds a matching membership for that family_id.
- **Frontend:** Initialize the Jetpack Compose Android app. Build out the design layout, styling rules, state management structures, and craft the Login, Registration, and empty Dashboard UI screens based on your Figma blueprints.

Sprint 3: The Private Family Feed (Week 3)

- **Backend:** Create CRUD API endpoints for publishing posts, fetching feeds, and logging simple interaction/reaction updates within a specific family room context.
- **Frontend:** Build out the interactive Jetpack Compose components for the family timeline, a clean photo/text upload modal, and large tapping targets for leaving reactions on family posts. Connect these components directly to your backend APIs.

Sprint 4: Community Notice Integration & E2E Testing (Week 4)

- **Backend & Database:** Build the public community notice board tables and endpoints, applying filters to fetch data strictly by geographic commune or district keys.
- **Frontend:** Integrate the workspace toggle switcher, allowing a user to easily flip their dashboard view between their private "Krousa Me" space and the public Local Community Board.
- **Verification:** Run extensive end-to-end multi-device local network testing to confirm that private circles are fully isolated from unauthorized accounts and that data loads cleanly across your entire V1 MVP architecture.